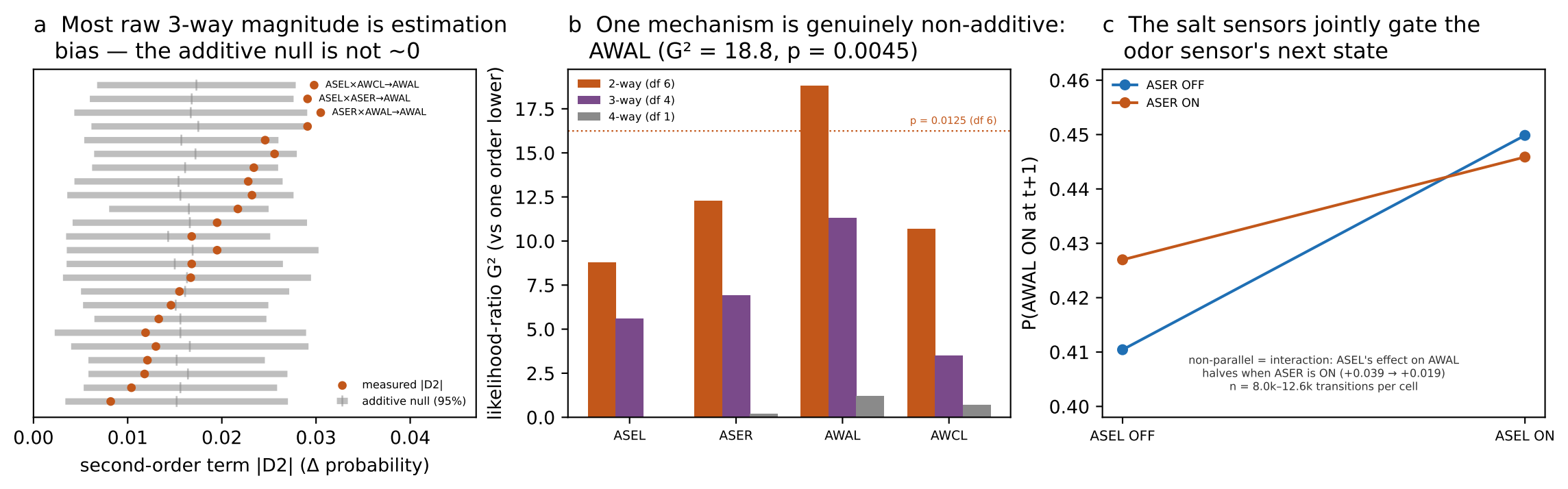
a Most raw 3-way magnitude is estimation bias — the additive null is not ~0
ASEL×AWCL→AWAL
ASEL×ASER→AWAL
ASER×AWAL→AWAL
measured |D2|
additive null (95%)
0.00
0.01
0.02
0.03
0.04
second-order term |D2| (Δ probability)
b One mechanism is genuinely non-additive: AWAL (G² = 18.8, p = 0.0045)
2-way (df 6)
3-way (df 4)
4-way (df 1)
p = 0.0125 (df 6)
likelihood-ratio G² (vs one order lower)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
ASEL
ASER
AWAL
AWCL
c The salt sensors jointly gate the odor sensor's next state
ASER OFF
ASER ON
P(AWAL ON at t+1)
0.40
0.41
0.42
0.43
0.44
0.45
0.46
non-parallel = interaction: ASEL's effect on AWAL
halves when ASER is ON (+0.039 → +0.019)
n = 8.0k–12.6k transitions per cell
ASEL OFF
ASEL ON